

Evidence-Grounded Auditing of Identification Assumptions in Climate-Policy Causal Evaluations

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Abstract

Difference-in-differences (DID) studies are widely used to evaluate climate policy, but assessing the evidence supporting their identification assumptions remains challenging. We introduce ARGUS, a structured language-model pipeline that audits reported evidence against an eleven-dimension assumption-implication-evidence rubric and abstains when relevant evidence cannot be retrieved. We evaluate ARGUS using injected flaws, economics papers, and a small pilot with reconciled labels. On the 11-flaw benchmark, ARGUS detects 73% of planted flaws, compared with 18% for a keyword-based pipeline. Across 26 economics papers, ARGUS abstains on about 40% of paper-dimension assessments for lack of retrievable evidence. In a five-paper pilot with labels reconciled by two annotators, it assigns a higher risk level than the labels on 25 of the 33 assessments it completes. A rule fixed before the labels arrived removes most of this in-sample; weighted agreement stays low. ARGUS provides evidence-linked risk reports that localize potential weaknesses for expert review, without adjudicating causal claims. Code and data: <https://github.com/yonghongzhang-io/ARGUS>.

1 Introduction

Difference-in-differences (DID) is widely used to evaluate climate and environmental policy, including carbon pricing, emissions-trading schemes, and emission standards. The credibility of these estimates depends on identification assumptions such as parallel trends, no anticipation, appropriate treatment of staggered adoption, and limited interference between units, each of which a large body of methodological work has shown to be fragile (Bertrand et al., 2004; Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; de Chaisemartin and

D’Haultfœuille, 2020; Roth et al., 2023). These assumptions can be hard to assess in climate settings, where policies may overlap, be phased in over time and spill across administrative boundaries, and outcomes may be modelled or remotely sensed (Appendix A). Where such estimates inform policy, causal conclusions built on weakly supported assumptions can mislead.

Assessing an identifying assumption requires more than locating a reported diagnostic. A paper may present parallel trends over an event-study figure, leaving the reader to weigh the design, the pre-period estimates, the inference, and the policy timing before accepting the claim; the assessment has to be repeated for every assumption and every paper. This is one facet of a broader concern with the reliability of empirical findings (Ioannidis, 2005; Silberzahn et al., 2018; Brodeur et al., 2020).

We argue that identification credibility is auditable even when the true causal effect is not, building on the research-design tradition in applied econometrics (Angrist and Pischke, 2010; Athey and Imbens, 2017; Rambachan and Roth, 2023). Because the counterfactual is unobserved, ARGUS (named after Argus Panoptes, the hundred-eyed watchman) does not ask whether an estimated effect is true. Instead, it asks whether each identifying assumption is adequately supported by the evidence reported in the paper, and where that support is weakest. ARGUS represents a DID study through an 11-dimension *assumption-implication-evidence* rubric and uses a bounded, retrieval-gated pipeline built on a large language model (LLM) to retrieve relevant evidence, assess its adequacy, and produce an evidence-grounded risk report for expert review (Figure 1). Model agency is confined to scoped retrieval and assessment tasks behind a fixed, deterministic control flow, unlike free-roaming agents whose control flow is model-decided (Yao et al.,

2023); ARGUS flags potential weaknesses rather than adjudicating causal truth.

Evaluating such an auditor is difficult because the true policy effect is unobserved. We therefore use *flaw injection* as local ground truth: starting from a supported study fixture, we introduce one known identification flaw and test whether ARGUS detects it, remains quiet on the clean version, and localizes the affected dimension. We complement this controlled evaluation with a run over 26 corpus papers tagged DID and a small diagnostic pilot with reconciled labels, used to identify calibration and applicability failures. Climate-policy evaluation motivates the rubric; the present evaluation uses synthetic environmental-policy fixtures and a general-economics corpus, so performance on a dedicated climate-policy corpus is untested.

Contributions. We develop and evaluate a rubric and pipeline for assessing the reported evidence behind DID identification assumptions, without access to the causal effect:

1. an *operationalization*: a machine-auditable *assumption–implication–evidence* rubric and a bounded, evidence-grounded pipeline with explicit abstention (§3);
2. an *evaluation methodology*: planted identification errors as local ground truth, with compute-graded and oracle ablations that locate the gain in per-dimension prompting and trace every synthetic omission miss to a gate abstention (§5);
3. a *diagnosis*: on real papers, grounding limits *coverage* while over-severity limits *agreement*, and a pre-specified rule derived on the same papers reduces the latter in-sample.

2 Related Work

DID identification and its threats. A large econometrics literature formalizes when DID estimators fail under staggered timing and heterogeneous effects: two-way fixed effects’ use of already-treated units as controls (Goodman-Bacon, 2021), negative weights (de Chaisemartin and D’Haultfoeuille, 2020), robust estimators (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021), assumption and pre-trend surveys (Roth et al., 2023), and fragile inference under serial correlation (Bertrand et al., 2004). ARGUS adds no estimator: it *operationalizes* this literature as an auditable rubric, turning each known

threat into a dimension whose supporting evidence can be checked in a given paper (§3).

LLM causal reasoning and benchmarks. A parallel line asks whether LLMs reason about causality. Benchmarks (CLadder (Jin et al., 2023), Corr2Cause (Jin et al., 2024), CausalBN-Bench (Zhou et al., 2026)) and analyses (Kıcıman et al., 2024; Yang et al., 2024; Zečević et al., 2023) report mixed evidence: strong performance on some causal tasks, but models often reproduce causal *language* without reliable causal *inference*; a few move toward applied settings and toward separating *identification* from *estimation* (Lee et al., 2025; Sawarni et al., 2026). ARGUS addresses a different task: whether a *human* paper’s identification assumptions are adequately evidenced. A causal verdict is out of scope by design.

Responsible AI for empirical research. Artificial intelligence (AI) is increasingly applied to research reliability: reproducibility checking, statistical-reporting validation, and AI-assisted review, motivated by wide analyst-driven result spreads (Silberzahn et al., 2018; Botvinik-Nezer et al., 2020) and specification searching in causal economics (Brodeur et al., 2020). LLMs are also used directly as evaluators and reviewers, judging model outputs (Zheng et al., 2023), giving feedback on research papers (Liang et al., 2024), and reviewing inside automated-science pipelines (Lu et al., 2024); these evaluators already produce ratings, pairwise preferences, and structured reviews. ARGUS applies an assumption-specific rubric to DID studies, links each dimension to in-paper evidence, and is measured against injected ground truth. ARGUS focuses on the evidence for *identification credibility* specifically (the assumptions supporting the causal interpretation) and issues no verdicts, producing evidence-grounded, human-reviewable reports that localize risk for the expert’s final call. Closest to our setting, CausalVerify (Zhang et al., 2026a) grounds evaluation in *execution*: it runs a model’s generated estimation code against synthetic data and checks the recovered effect. That benchmark does not audit identification assumptions, which is the layer ARGUS takes up; we reuse its released corpus of economics papers and add a dimension-level identification-risk annotation layer on top of it.

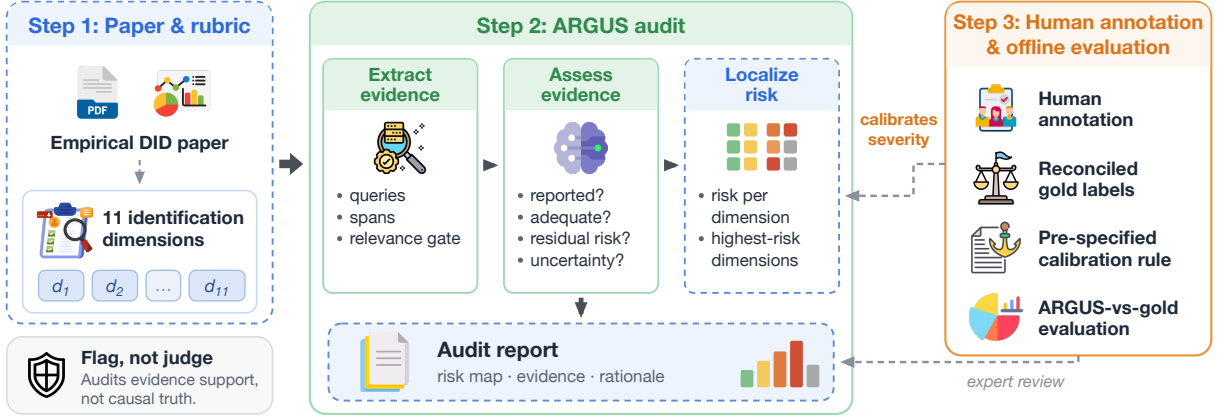


Figure 1: Overview of ARGUS. Step 1 converts an empirical DID paper into an auditable structure using an 11-dimension identification rubric. Step 2 retrieves dimension-specific evidence, assesses reported-evidence adequacy with a bounded LLM audit module, and localizes risk in a human-reviewable report. Step 3 uses labels from two annotators, reconciled by them, to evaluate audit severity and score a pre-specified calibration rule offline; deployment requires only expert review of the report. Colours as in Figure 2: blue, deterministic steps; green, model calls; orange, human steps.

Natural language processing for climate evidence. Natural language processing (NLP) systems for climate text verify claims against evidence (CLIMATE-FEVER (Diggelmann et al., 2020)), detect environmental claims (Stammbach et al., 2023), analyse corporate climate disclosures with traceable LLM answers (Ni et al., 2023), and tune models for faithful evidence-based question answering (Schimanski et al., 2024). These systems ground answers in documents; ARGUS extends document-grounded verification from *claims* to the *identification argument* of a causal study, auditing whether reported evidence supports each assumption.

Evaluation methodology and bounded agents. Our evaluation follows benchmarking lessons: holistic evaluation (Liang et al., 2023) over a single number; dynamic, adversarial benchmarking (Kiela et al., 2021) (which inspires our controlled flaw injection); and the caution that operationalizing a capability is non-neutral (Raji et al., 2021; Gebru et al., 2021) (§6). A closely related concern is that an agent benchmark can measure its harness rather than its model: Zhang et al. (2026b) show that a fixed scaffold making the execution-critical decisions and a scorer grading output shape mask each other, so repairing either alone leaves the ranking uninformative. Our ablations apply a related check inward: they compare joint and per-dimension prompting, with and without retrieval and relevance gating, holding the model and rubric fixed (§5.2). Where ReAct-style

agents (Yao et al., 2023) choose their own action sequence, ARGUS fixes the order of retrieval and assessment and confines model calls to those two stages, trading autonomy for inspectability and evidence-traceability.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions of identification, design, inference, and reporting: parallel trends, no anticipation, staggered-timing handling, SUTVA (stable unit treatment value assumption, no interference) and spillovers, control-group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* or design requirement (what must hold), a testable *implication* (the diagnostic that should then be observable), and the *evidence* a credible paper would report to support it. For a carbon-pricing or emissions-trading evaluation, for instance, the parallel-trends dimension expects an event-study or pre-trend test with pre-period estimates near zero (an insignificant pre-trend alone can reflect low power; Roth, 2022), while no-anticipation turns on the gap between a scheme’s *announcement* and *implementation* dates. The rubric is declarative, stored in a configuration file (`identification_dimensions.yaml`), so the audited dimensions are explicit and inspectable rather than buried in code. A compan-

ion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded LLM pipeline

The audit runs a fixed, deterministic sequence, *decomposition* → *extraction* → *assessment* → *localization* → *report*, shown in the top band of Figure 2. The stage order is written in code and never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand; *extraction* gathers, per dimension, the in-paper evidence relevant to that assumption; *assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence; *localization* aggregates these judgements into a risk map that ranks the dimensions assigned the highest risk (ties broken by the fixed rubric order); and *report* renders a transparent, evidence-grounded document for a human reviewer.

Model calls are confined to two stages, extraction and assessment. In the configuration evaluated in §5 they resolve, for each dimension, to a fixed sequence: deterministic lexical retrieval over the paper’s sections and captions, one relevance-gate call that labels each retrieved passage *high*, *partial*, or *none*, and, if at least one passage is kept, one adequacy call. If no passage is kept the dimension is scored *unknown* without an adequacy call; a dimension with only partially relevant passages is still judged (Appendix D). Figures and tables enter only through their captions. The code also defines a more general bounded loop for these two stages (a fixed tool set of `evidence_search`, `figure_parse`, and `policy_lookup`; a hard step budget, `max_steps`; a logged trace of every step and cited passage); the reported experiments run it with `max_steps=1` and never invoke `figure_parse` or `policy_lookup`, and all other stages are plain orchestration with no model call. We therefore call the design *bounded* rather than *agentic*: the evaluated system is structured, dimension-scoped prompting over retrieved evidence with an explicit abstention path, not an open-ended agent.

Why bound the model’s role. Fixing the stage order makes the sequence of operations explicit and comparable across runs and keeps every judgement anchored to logged evidence a human

can inspect. It does not make model outputs identical: Appendix F reports the variation observed across repeated runs. The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk; the judgement itself stays with the human.

3.3 Output

ARGUS emits study–dimension–risk triplets, each with an evidence-grounded rationale, alongside the human-readable report. In the evaluated prompts (Appendix E) *low*, *medium*, and *high* correspond approximately to retrieved evidence judged sufficient, partial, and missing or flawed; *unknown* is produced by the gate when no relevant passage is retained and reflects retrieval, not a judgement about the paper. None of the four is a verdict on the causal effect. This output shape keeps the system both auditable by a human and measurable by the evaluation that follows.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest (the true causal effect of a carbon price, an emissions cap, or an emission standard) is never observed. We therefore do not attempt to certify that ARGUS recovers true effects. Instead we construct *local* ground truth by controlled perturbation. Beginning from a short synthetic fixture, an environmental-policy DID study that reports supporting evidence on all eleven dimensions, we inject exactly one known flaw that targets a single dimension, drawn from the flaw taxonomy (`flaw_taxonomy.yaml`; Appendix B traces each flaw to the literature that names it). The target dimension is known by construction, and the expected response (risk of at least *medium* on that dimension) is defined by the rubric, without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension to *medium* or above? (All reported results use this threshold.)

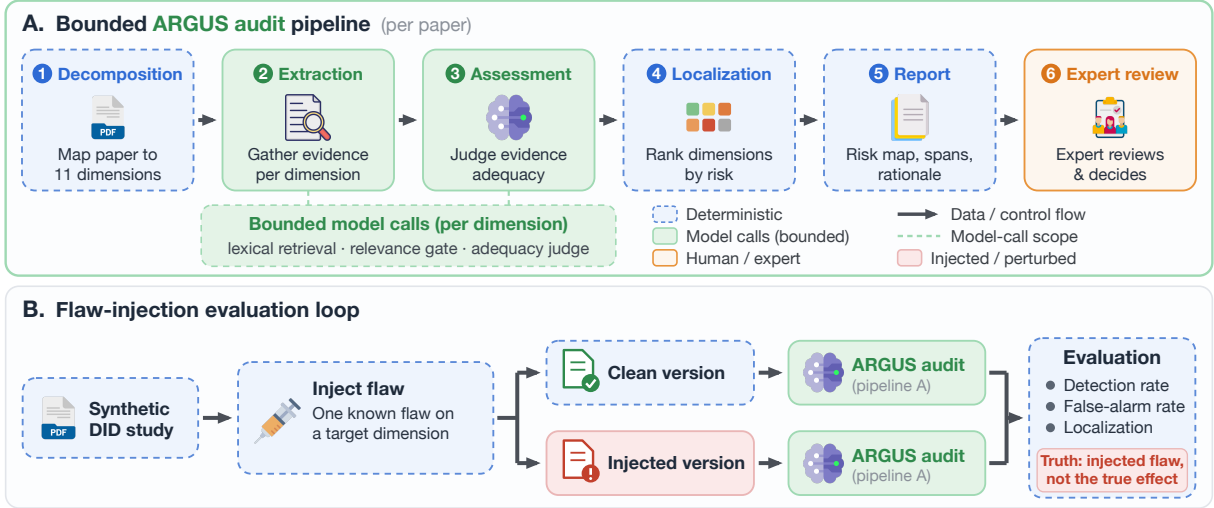


Figure 2: ARGUS architecture. The top band is the audit pipeline, a fixed deterministic sequence; model calls are confined to two stages (extraction and assessment). As evaluated, these resolve per dimension to a fixed call sequence (lexical retrieval, one relevance-gate call, one adequacy call) with a step budget of one and a logged trace; the designed tools `figure_parse` and `policy_lookup` are not exercised (§3.2). The bottom band shows the flaw-injection evaluation loop, which injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth, never against the unobserved true effect.

- **False alarm:** does the clean version stay quiet on that dimension?
- **Localization:** is the targeted dimension the one the auditor flags most strongly?

detector looks for. A regression test fails if an injected paper contains the target dimension’s predefined negative-signal phrases, so this phrase leakage cannot silently return; detection cannot come from matching those phrases.

4.2 Structural perturbation, not sentinels

How a flaw is injected is a methodological choice that matters. A naïve injector appends a sentence that *names* the flaw: e.g. “treated and control have non-zero pre-period coefficients with significant leads.” If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector matches on. In our initial implementation this produced a detection rate of 1.000 inflated by phrase leakage.

We replace this with *structural perturbation* of two kinds. *Omission* flaws *remove* the section, figure, or table that supplies the evidence the rubric specifies for a dimension (e.g. deleting the placebo-robustness section or the balance table). They test sensitivity to missing reported support; they do not establish that the underlying assumption is violated. *Commission* flaws *rewrite* a section to introduce a stated design problem in prose intended to resemble how a flawed paper would describe it (e.g. pooling all province-years in a two-way fixed-effects regression under staggered policy timing), *without* the phrases the keyword

5 Results

We evaluate ARGUS along four axes: a synthetic flaw-injection protocol, an architecture ablation, a run over 26 papers tagged DID in an economics corpus, which stress-tests the audit architecture across empirical economics beyond the climate-policy setting that defines its target deployment, and a pilot with reconciled labels. Four findings emerge: (i) LLM pipelines that judge evidence *adequacy* detect planted flaws that a keyword-presence pipeline misses; (ii) per-dimension prompting detects more planted flaws than a single joint call, while the retrieval gate turns retrieval failure into an auditable *unknown* signal; (iii) on real papers retrieval limits coverage, and weakly relevant passages are associated with high-risk ratings; and (iv) against the reconciled labels ARGUS is systematically over-severe, a bias that a pre-specified rule derived on the same papers reduces in-sample.

Table 1: Synthetic flaw injection on the `clean_supported` fixture, eleven flaws; all LLM rows gpt-4o, detection replicated on the dated 2024-11-20 snapshot; see Appendix F on run-to-run differences. Every row is an end-to-end pipeline with its own evidence path (keyword and two-stage do not share retrieved evidence, Appendix D; Appendix K holds it fixed); the ablation arms share the model and rubric but remove retrieval, the gate, or both. The final row is the leaked, pre-fix circular setup. On the 33-variant benchmark the two-stage assessor rates one clean dimension medium, a 0.09 over-flag (§5.2). Wilson 95% intervals for the main rates are in Table 8.

assessor	calls	detection	false alarm	localization
keyword baseline	0	0.182	0.000	0.182
single-pass	1	0.818	0.000	0.818
per-dimension, no retrieval	11	1.000	0.000	1.000
two-stage ARGUS	22	0.727	0.000	0.727
(circular, pre-fix)	—	1.000	0.000	1.000

5.1 Adequacy pipelines versus keyword presence; what the gate trades

We compare three assessors under one flaw-injection protocol on the `clean_supported` fixture (one flaw injected per run, all eleven flaws; `max_steps=1`). A deterministic **keyword baseline** scores each dimension from the presence of supporting and risk-indicating vocabulary. The **two-stage ARGUS assessor** judges evidence *adequacy* (whether the reported evidence supports the assumption under the rubric). The two are complete pipelines with their own evidence paths: the baseline scores paragraph chunks, while the two-stage path re-retrieves and gates whole sections (Appendix D). The pair is therefore an *end-to-end* comparison and cannot separate the judgement policy from retrieval. A **single-pass** ablation removes the architecture entirely: the full paper and all eleven dimensions in one gpt-4o call, with the same model, temperature (the sampling-randomness setting, held at 0), risk schema, and scoring harness, but no staged retrieval, relevance gating, or bounded tool use; its strict output schema admits only low, medium, or high, so this arm cannot abstain (Appendix E). Table 1 reports the headline metrics.

The keyword baseline is blind to commission flaws. The baseline never raises a false alarm and correctly flags the two omission flaws (`measurement_break` and `spillover_contamination`) by the *absence* of their vocabulary, but beyond those detection is only 0.182 (Figure 3b). Two failure

modes explain it: the same keywords occur in other sections (deleting the placebo section does not help when another section also mentions “placebo”) and blindness to *every* commission flaw, whose flawed-but-plausible evidence carries the expected topic vocabulary.

Detection improves with the LLM pipelines.

Replacing the keyword pipeline with the two-stage LLM pipeline raises detection from 0.182 to 0.727 with no false alarms, catching *every* commission flaw the keyword scorer is blind to; handing each dimension the full fixture (the per-dimension arm) reaches 1.000. Since retrieval also differs, a pre-specified control holds the evidence fixed (Appendix K). Reading the keyword pipeline’s own evidence chunks, with no gate, the adequacy judge detects 10 of 11 flaws against the keyword scorer’s 2 (exact McNemar $p=0.008$), and 31 against 8 of the 33 variants. The gain is a trade: the judge rates one clean dimension medium and 11–16% of non-target dimensions on injected papers, where the keyword scorer flags none. The model reads flawed-but-natural prose the keyword scorer cannot; the audit report in Appendix I shows it flagging the injected anticipation flaw with the announcement-gap rationale, with no sentinel keyword present.

What the gate trades. The gated pipeline detects *less* than its own ablations here (0.727 vs 0.818 and 1.000) by design: its three misses are all *omission* flaws on which retrieval finds nothing and it abstains rather than guessing; in the reported runs each of these misses terminates at the relevance gate as `unknown`, and the oracle-retrieval arm of §5.2 flags the same cases when the assessor is called on the target section. The gate’s effect is behavioural and shows on real papers. A single-pass variant whose prompt *does* permit `unknown` (“use sparingly”; Appendix E) abstained on one of 275 cells (0.4%), against 38.9% mechanical abstentions for the gated pipeline (Appendix C). The designs behave differently when evidence is hard to ground; without cell-level gold the contrast does not show which is more reliable.

Scope. These are short synthetic fixtures; ceilings on clearly injected flaws are a proof-of-concept, tested next on 33 variants (§5.2) and real papers (§5.3).

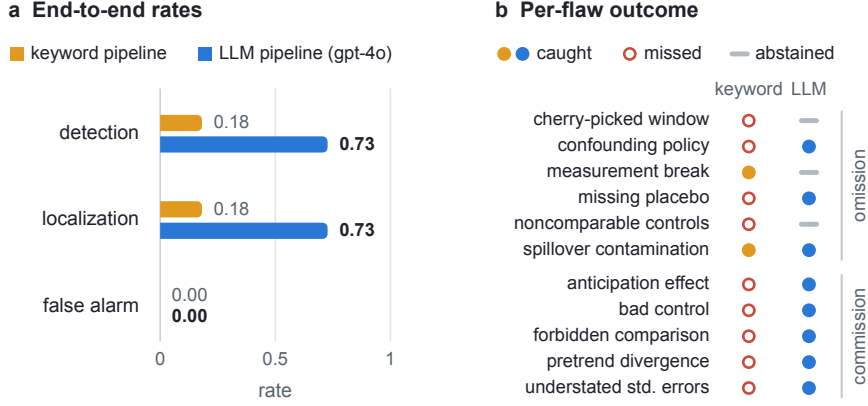


Figure 3: Keyword pipeline vs LLM evidence-adequacy pipeline ($n=11$ flaws, structural injection). **(a)** End to end, the LLM pipeline raises detection and localization $0.182 \rightarrow 0.727$ with false alarm 0 (the two pipelines retrieve evidence differently; Appendix K holds it fixed). **(b)** Per-flaw caught/missed: the keyword baseline is *blind to every commission-type flaw*, all of which the LLM catches; the LLM’s three misses are omissions on which it abstains. Table 2 reports the 33-variant benchmark.

5.2 An expanded benchmark of 33 flaw variants

The eleven flaws are deliberately clear-cut, so we built an expanded benchmark: 33 variants, three per dimension (two *commission* rewrites, one *omission*). The rewrites were written to resemble descriptions of flawed design choices and are verified free of the keyword detector’s predefined negative-signal phrases. Three two-stage runs (gpt-4o, temperature 0) are near-deterministic (detection identical on 32/33 variants, risk labels on 31/33; Appendix F). Detection is 0.73–0.76 and localization 0.64–0.70, in line with its behaviour on the eleven clear flaws.

Where the gain comes from. A compute-upgraded ablation on this benchmark (Appendix C) removes retrieval and gating in turn, the de-scaffolding check of Zhang et al. (2026b) applied to our own pipeline: a full-paper single pass (one call) detects 0.727; per-dimension prompting without retrieval (eleven calls) detects 0.879; the full two-stage pipeline (twenty-two calls) detects 0.747. Per-dimension prompting detects more variants than a single joint call (29/33 vs 24/33); the comparison changes both the task decomposition and the number of calls, so it does not isolate their separate contributions. The retrieval-and-gate layer then *trades* omission detection for abstention (on these short fixtures a gateless assessor can tell from its context that the section is missing; on real papers it commits judgements nearly everywhere, §5.3). An *oracle-retrieval* arm supplies a diagnostic comparison: fed the target

section directly (empty for omissions, bypassing the gate), the assessor detects 0.86 of commissions and 1.00 of omissions, while the gate suppresses the alarms section-level evidence alone triggers (0.21 oracle arm vs 0.09; the full-context arm’s 0.00 reflects its wider view). At $n=33$ none of these pairwise arm differences reaches significance under an exact paired McNemar test (single-pass vs per-dimension $p=0.13$, per-dimension vs two-stage $p=0.22$, two-stage vs oracle $p=0.13$; Appendix G); the claim concerns the observed execution path (every extra omission the oracle arm catches is a case the gated pipeline had scored *unknown* at the gate), not a significant rate gap.

Cross-model robustness. Porting the unchanged pipeline across providers shows that high commission detection is not model-specific (Appendix C): *commission* detection is high everywhere (gpt-4o 0.89, Claude Opus 4.8 1.00, Gemini 2.5 Flash 0.91, Llama 3.1 8B 0.95), but the clean-fixture flag rates differ substantially (Table 5), so high detection alone does not show that each model separates the flawed and clean versions reliably. What differs is the operating point, and Table 6 shows where: Opus 4.8 rates the *clean* fixture medium or high on 18 of 22 commission pairs (false alarm 0.82), the over-severity the human gold finds for gpt-4o, only stronger; Llama 8B almost never abstains (one *unknown* in eleven omissions) and calls the missing evidence high; Gemini is closest to gpt-4o (26/33 identical injected verdicts). High commission detection is observed with every tested model; the abstention

Table 2: The 33-variant benchmark (gpt-4o, mean of three runs at temperature 0; detection decisions identical on 32/33 variants). Adequacy reasoning catches *commission* flaws; *omission* misses are abstentions to *unknown*, marking the evidence-grounding bottleneck rather than wrong judgements. Intervals in Table 8.

variant type	n	detection	localization
commission	22	0.89	0.82
omission	11	0.45	0.33
all	33	0.75	0.66

behaviour and the false-alarm profile are not, so changing the model requires a fresh assessment of calibration and false alarms (§5.6).

Detection differs by flaw type (Table 2). The assessor catches 0.86–0.91 of *commission* flaws (flawed-but-present evidence, the case adequacy reasoning targets) but only 0.45 of *omission* flaws. The omission misses are not confident errors: the six missed variants all abstain to *unknown* rather than assert low risk, because deleting the supporting section makes retrieval fail. This is the same failure point we find on real papers (§5.3): on these fixtures the assessor flags most commission variants when the evidence is retrieved and abstains when it is not. The only false-alarm signal (0.09 in every run) is one clean-fixture dimension rated medium (*inference* in run 1, *specification* in runs 2 and 3), the same over-severity we quantify against the human gold (§5.5); the clean audit is itself not deterministic. Table 1 shows no false alarm on the original eleven-flaw set.

5.3 Real papers: retrieval limits audit coverage

We next audit the 26 papers tagged difference-in-differences in the CausalVerify corpus (Zhang et al., 2026a) (Appendix H gives its composition and the limits of that tag). There is no injected ground truth; we report the raw per-dimension risk distribution of each assessor (Figure 5). The keyword baseline is permissive: 91% of its judgements are *low*. It sees topic vocabulary, declares the dimension supported, and flags almost nothing. The two-stage LLM assigns a wider range of risk labels, and its most informative output is that $\sim 40\%$ of judgements are *unknown*: the relevance gate could not surface evidence, so the system says so rather than over-flagging. Two regimes appear: dimensions whose discussion is diffuse or lives in figures the text retrieval cannot

reach are dominated by *unknown*, while dimensions where a section is retrieved are judged inadequate. On real papers, then, retrieval limits coverage, and weakly relevant passages are associated with high-risk ratings. These results identify retrieval and the treatment of missing evidence as failure points; they do not separate their contributions from rubric applicability and assessment errors. The design therefore reports retrieval quality and an explicit *unknown* state distinct from a substantive high-risk judgement.

Why fixtures abstain but real papers flag. The synthetic omission misses of §5.2 become abstentions, yet on real papers a missing check usually becomes a substantive *high*. The asymmetry is one branch of the gate. Retrieval is lexical (Appendix D): deleting a section from the short fixture leaves no passage that even shares its vocabulary, the gate finds nothing relevant, and the pipeline abstains without calling the judge. A full paper usually contains a section that shares the vocabulary, so the gate returns *partial* relevance, the judge is called on an off-target passage, reports the evidence as missing, and the prompt maps missing to high. The pilot shows the pattern (Table 7): all 22 failed-retrieval cells are *unknown*, and 20 of the 22 weak-retrieval cells are *high*, 19 of them with *evidence_status=missing*. This is the branch the calibration layer of §5.6 targets.

Judgements cite specific evidence (Appendix I gives a qualitative example). Roughly 44% of judgements on these papers are *high*; without dimension-level human labels beyond the five pilot papers we cannot say whether this is warranted or over-strict where retrieval succeeds.

5.4 An annotation pilot with reconciled labels

To test agreement with human judgement, we built a small reconciled reference set (the “gold”): two annotators from outside the author team, who had no part in building ARGUS, independently labelled five corpus papers $\times$ 11 dimensions (55 cells), without AI tools or sight of ARGUS’s output, and then reconciled the 13 cells on which they differed in applicability or risk between themselves, without the first author (Appendix J). Their independent labels agree on 45 of the 54 cells both rated (Cohen’s $\kappa = 0.56$; quadratic-weighted $\kappa = 0.62$), every disagreement is one severity step, and 23 of the 55 cells are labelled under an *analogue* reading of the dimension because the de-

sign is not a canonical DID. The reconciled gold (55 cells) is low 7, medium 46, high 2; we read this pilot as preliminary. The reviewed version used labels two people had produced with LLM assistance and sent to the first author, with no annotation record; they are replaced here (agreement with the human gold 31/55, Appendix J).

5.5 ARGUS versus the human gold

We use this gold as a *diagnostic pilot*, not a validation of accuracy: it localizes *how* ARGUS fails (Table 13, Figure 4). ARGUS abstains on 22/55 cells, and when it answers its errors are almost entirely one-directional: 25/33 answered cells are more severe than the reconciled gold, none less severe; the gold contains two high-risk cells, whereas ARGUS assigns 24. With so few high gold labels, precision and recall are noisy (of the two high gold cells ARGUS flags one and abstains on the other). Most abstentions are retrieval failures rather than absent evidence: both annotators located reported evidence on 19 of the 22 abstained cells, at least one on 21.

5.6 Calibration: a pre-specified rule, in-sample

Of the re-run’s 24 over-severe *high* judgements, 23/24 have `evidence_status=missing` and 20/24 have `retrieval_quality=weak`. ARGUS treats “weak retrieval found nothing” as a substantive *high*, conflating a *retrieval failure* with the paper lacking the evidence, which the reconciled human label puts at low or medium: a fixable calibration target. The pattern re-emerges in every leave-one-paper-out fold of the pilot (a recurrence check; no rule is refitted). (This uses a re-run with retrieval-quality and evidence-status fields; it differs from the corpus run of §5.5 in one cell; its pre-calibration figures are the same 25/33 over-severe and exact agreement 0.24.)

A deterministic rule over fields the LLM already produces reduces the over-severity: a weak-retrieval *high* (20 cells) is demoted to *medium* (keep coverage) or to *unknown* (abstain). Keeping coverage, the rule raises exact agreement from 0.24 to 0.76 (8/33 to 25/33) and cuts over-severe cells from 25 to 8: on the 33 answered cells it changes seventeen incorrect labels to the reconciled label and no correct label to an incorrect one, and keeps the one high gold cell ARGUS had caught. The abstention variant removes those 20 predictions (coverage 33/55 $\rightarrow$ 13/55) and leaves

the remaining labels unchanged. Three further single-cell rules lift exact agreement to 0.85 without raising weighted κ under the demotion policy; they are post-hoc string and field rules, reported as exploratory (Appendix J, Table 14, Figure 6). Exact agreement is a weak yardstick here: the gold is 84% *medium*, and a constant *medium* label would score 26/33 on the same cells; the statistic to read is weighted κ , $0.08 \rightarrow 0.23$, with an interval that includes zero (Appendix G). The rules were fixed before the human labels arrived but were derived on these same five papers against the earlier labels, so the lift is in-sample: it points the same way in all five papers and is untested on unseen papers.

6 Discussion

What ARGUS measures. A measurement is only as good as the link between the construct and the task that operationalizes it (Jacobs and Wallach, 2021; Raji et al., 2021). ARGUS audits whether a climate-policy study’s identification assumptions are *adequately supported by its reported evidence*: reported-evidence adequacy is an operational proxy for identification credibility, not causal validity itself.

Why climate policy. Climate-policy estimates inform governance, so a confidently wrong credibility judgement has a cost; ARGUS therefore abstains when its gate finds no relevant passage.

Retrieval limits audit coverage. A further gap is retrieving the relevant evidence. On the real-paper corpus it limits coverage, and in the pilot an assessor handed weakly relevant evidence usually rates the dimension high (20 of 22 weak-retrieval pilot cells), so ARGUS reports retrieval quality and an explicit *unknown* distinct from a substantive high-risk judgement.

7 Conclusion

The causal effect of a climate policy is unobservable; whether a study’s reported evidence supports its identification assumptions can be assessed. ARGUS operationalizes that assessment with a bounded, evidence-grounded pipeline, flaw injection as local ground truth, and a reconciled pilot whose human labels evaluate the auditor and score a pre-specified rule that reduces its over-severity in-sample; its value in expert review and on climate-policy papers is untested.

Limitations

Our human gold is pilot-scale: 5 papers and 55 cells from two annotators who reconciled their differences themselves, so the comparison is a preliminary signal rather than a calibrated benchmark; its labels are 84% *medium*, which makes exact agreement lenient and κ the more informative statistic. The gold is the reconciled judgement of two annotators, not of a panel of DID methodologists; it has only two *high* cells; 23 of its 55 cells judge an analogue of the assumption because the design is not a canonical DID; the main ARGUS-versus-gold figures are conditional on the 33 cells ARGUS answered (it abstained on 22, on 19 of which both annotators found evidence); and a constant *medium* label would outscore the calibrated system on exact agreement. The current scope is confined to difference-in-differences identification; whether the rubric-plus-injection methodology transfers to other quasi-experimental designs is untested. The rubric, flaw taxonomy, and injector are co-designed, so synthetic detection rates are best read as internal consistency under known threats rather than external generalization (Appendix B traces each injected flaw to the literature that names it, so the threat catalogue is inherited while the injector’s prose is ours); flaws authored by independent experts are future work. The benchmarks are small: at $n=11$ and $n=33$ no pairwise difference between architecture arms reaches significance under a paired exact test, and the human-gold intervals are wide (Appendix G), so we report descriptive differences on these benchmarks; the samples do not establish population-level differences between the architecture variants. Those intervals and the cell-level paired tests also treat the 55 pilot cells as independent although they are nested in five papers, and the calibration rules were derived on the pilot they are scored on, so the calibration lift is an in-sample result whose size on unseen papers is untested. The parallel-trends criterion is likewise a reporting check: ARGUS does not test whether a pre-trend test had the power to detect a violation. The real corpus is a general-economics stress test that contains no climate-policy evaluation in the narrow sense (Appendix H); a climate-policy corpus run is the natural next test. Relatedly, flaw injection certifies the auditor against the *injected* threat on a single dimension only and inherits the taxonomy’s blind spots, so we treat detection and local-

ization as necessary rather than sufficient, pairing them with the false-alarm rate and the human-gold comparison. Evidence grounding limits coverage. ARGUS emits *unknown* only when the relevance gate finds no suitable passage; partial matches still reach the judge and can produce over-severe verdicts. Finally, ARGUS screens whether reported evidence supports a study’s identification assumptions and does not adjudicate the true causal effect, which is never observed. Its outputs surface risk for expert review and do not replace it.

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A Domain Motivation and Design Rationale

A.1 Why climate-policy DID is a demanding audit setting

Climate-policy evaluations can raise several identification challenges at once. Policy announcement, legal adoption, and actual implementation may occur at different dates, creating scope for anticipation. Carbon markets and environmental regulations may be introduced at different times across jurisdictions, so that heterogeneous and staggered treatment effects need to be handled. Emissions, investment, and production responses can spill across regional borders, challenging no-interference assumptions. Climate policies can overlap with industrial, energy, and air-pollution regulations, complicating the exclusion of concurrent interventions. Policy effects may emerge with long and heterogeneous lags. Emissions and environmental outcomes may be assembled from administrative, modelled, or remotely sensed sources whose measurement properties vary across places and time. These features do not make DID unique to climate policy; they motivate a rubric that makes the evidence for each design choice explicit.

A.2 Identification auditing and research reliability

ARGUS addresses one component of a broader research-reliability problem: published conclusions can vary with specification choices, undisclosed analytic flexibility, and researcher degrees of freedom (Leamer, 1983; Ioannidis, 2005; Simmons et al., 2011; Gelman and Loken, 2014; Silberzahn et al., 2018; Brodeur et al., 2020; Botvinik-Nezer et al., 2020). This is

the identification-credibility component of the broader research-design or “credibility revolution” program in applied econometrics (Angrist and Pischke, 2010; Athey and Imbens, 2017), which increasingly treats an assumption like parallel trends as a matter of degree rather than a binary (Rambachan and Roth, 2023; Roth, 2022). ARGUS does not attempt to audit every component of an empirical paper; it focuses on the assumptions that support a causal interpretation.

A.3 Why the pipeline is bounded

Open-ended agents may choose different tools and reasoning paths across runs, making their outputs difficult to reproduce and compare. Large language models are increasingly used for evaluative and reviewing tasks (Zheng et al., 2023; Liang et al., 2024; Lu et al., 2024), but as open-ended agents their reasoning traces are hard to audit. ARGUS instead fixes the pipeline order in code and confines model-driven decisions to evidence retrieval and adequacy assessment. Each model call operates under a fixed tool set, a hard step budget, and a logged trace. This design trades autonomy for inspectability and controlled evaluation under flaw injection; it fixes the sequence of operations, not the model’s outputs.

B Flaw Taxonomy Provenance

The rubric, taxonomy, and injector are co-designed, but the catalogue of threats is not ours: each injected flaw is a threat named in the DID and research-design literature (Table 3). What we authored is the injector’s prose for each threat, which is why we describe synthetic detection as internal consistency under known threats and treat independently authored flaws as future work.

C Ablation Matrix and Cross-Model Panel

Table 4 grades compute on the 33-variant benchmark: per-dimension prompting (one adequacy call per dimension) detects more variants than a single joint call, and the retrieval-and-gate layer converts omission detection into abstention. Table 5 ports the unchanged pipeline across providers: commission detection is uniformly high, while false-alarm and localization profiles vary sharply by model. Cross-provider decoding defaults differ (newer reasoning models reject an explicit temperature), so these are single runs. Ta-

Table 3: Each injected flaw, its target dimension, and the literature that names the threat.

flaw	dimension	threat named in
pretrend divergence	parallel trends	(Roth, 2022; Roth et al., 2023; Rambachan and Roth, 2023)
anticipation effect	no anticipation	(Callaway and Sant’Anna, 2021; Roth et al., 2023)
forbidden comparison	staggered timing	(Goodman-Bacon, 2021; de Chaisemartin and D’Haultfoeuille, 2020; Sun and Abraham, 2021)
spillover contamination	SUTVA / spillovers	(Athey and Imbens, 2017; Roth et al., 2023)
noncomparable controls	control group	(Callaway and Sant’Anna, 2021; Athey and Imbens, 2017)
bad control	specification	(Angrist and Pischke, 2009)
understated standard errors	inference	(Bertrand et al., 2004)
cherry-picked window	sample period	(Simmons et al., 2011; Brodeur et al., 2020)
confounding policy	concurrent policies	(Meyer, 1995)
missing placebo	placebo / robustness	(Bertrand et al., 2004; Athey and Imbens, 2017)
measurement break	data measurement	(Meyer, 1995)

Table 4: Compute-graded ablation on the 33-variant benchmark (gpt-4o). det / fa / loc = detection, false alarm, localization. Calls per paper are the nominal maximum: when the gate finds nothing relevant the adequacy call is skipped, and retries are not counted. Localization takes the top-ranked flagged dimension; ties in risk level are broken by the fixed rubric order.

arm	calls/paper	det	fa	loc
full-paper single pass	1	0.727	0.000	0.727
per-dimension, no retrieval	11	0.879	0.000	0.879
two-stage (retrieval + gate)	22	0.747	0.091	0.657

ble 6 shows where the operating points diverge: how each model treats a deleted section, how often it already rates the unmodified fixture medium or high, and how often its injected-variant verdict matches gpt-4o’s.

On real papers the gate’s behavioural contrast sharpens. This run used a second single-pass implementation, distinct from the strict-schema arm of Tables 1 and 4: its prompt permits `unknown` with the instruction to “use sparingly” (Appendix E). Over the 25 papers common to both runs (275 cells) it chose `unknown` once (0.4%; a model-written abstention with a rationale, not a harness default), where the gated pipeline abstains mechanically on 38.9%. The gated pipeline’s `unknown` cells become low (1.8% → 14.9%) or medium (14.2% → 43.6%) judgements while the high-risk share stays comparable (gated 45.1%, single pass 41.1%). These shares describe behaviour (a discretionary abstention option is rarely taken, a mechanical one often is); without cell-level gold they do not show which judgements are more reliable.

Table 5: Cross-model panel on the 33-variant benchmark (single run per model; gpt-4o is the three-run mean; single runs support the qualitative porting claim only). comm/omit = commission/omission detection. Intervals in Table 8.

model	det	fa	loc	comm	omit
gpt-4o	0.75	0.09	0.66	0.89	0.45
Claude Opus 4.8	0.91	0.82	0.24	1.00	0.73
Gemini 2.5 Flash	0.79	0.18	0.36	0.91	0.55
Llama 3.1 8B (local)	0.94	0.27	0.27	0.95	0.91

Table 6: Cross-model operating points on the 33-variant benchmark (single runs; gpt-4o run 1). Omission column: verdicts on the eleven deleted-section variants, in the order unknown / high / medium / low. Clean column: commission pairs (of 22) whose *unmodified* fixture is already rated medium or high. Last column: exact agreement with gpt-4o’s injected-variant verdict.

model	omission verdicts	clean $\geq$ med	agree
gpt-4o	6 / 3 / 2 / 0	2 / 22	–
Claude Opus 4.8	3 / 7 / 1 / 0	18 / 22	20 / 33
Gemini 2.5 Flash	4 / 3 / 3 / 1	4 / 22	26 / 33
Llama 3.1 8B (local)	1 / 10 / 0 / 0	6 / 22	19 / 33

D Retrieval Details

The only corpus queried is the paper under audit; there is no external corpus and no embedding index. Papers are ingested through a light markdown convention (## section headings plus @figure/@table caption lines), and the searchable blocks are the title, abstract, each section, and each figure or table caption. On the LLM path retrieval is section-level and lexical: each dimension has a fixed query pack of 11–19 distinct terms (e.g. *pre-trend*, *event study*, *lead coefficients* for parallel trends), a block scores by the summed frequency of its matched terms, and the top $k=4$ blocks are kept, each truncated to 2,000 characters. The relevance gate sees the first 1,200 characters of each candidate and labels it *high*, *partial*, or *none*; retrieval quality is *good* if any candidate is high, *weak* if only partial candidates remain, and *failed* otherwise. A failed dimension is scored unknown without an adequacy call; weak and good dimensions pass the kept blocks, in full, to the adequacy judge together with the rubric’s assumption, implication, and expected evidence. The keyword baseline instead searches paragraph chunks capped at 900 characters (sentence-split when longer) with a single query string and top $k=5$, scoring positive and

Table 7: Retrieval quality versus emitted risk on the 55 pilot-gold cells (re-run recording retrieval quality and evidence status). Failed retrieval always abstains; weak retrieval almost always becomes a substantive *high*, 19 of those 20 with evidence_status=missing.

retrieval quality	unknown	high	medium	low
failed (22)	22	0	0	0
weak (22)	0	20	2	0
good (11)	0	5	5	1

negative sentinel phrases. With `max_steps=1` there is no iterative exploration, so breadth is the top- k cap and depth is one round. Chunk size, k , and embedding retrieval were not swept. The oracle-retrieval arm of §5.2 is a diagnostic comparison in which the target section is supplied directly and the gate is bypassed; it is not a formal upper bound on retrieval performance. Table 7 shows how the gate’s three outcomes map to risk on the pilot-gold cells, the mechanism behind the fixture/real-paper asymmetry of §5.3.

E Prompts

All gpt-4o calls run at temperature 0 (other providers: Appendix F); all but the real-paper single-pass variant at the end of this appendix use a strict JSON schema. Angle-bracketed fields are substituted from the rubric entry or the retrieved passages; nothing else varies across dimensions or papers. The per-dimension no-retrieval arm of §5.2 reuses the adequacy prompt below verbatim, passing the whole paper as the evidence items instead of retrieved sections: the adequacy prompt is unchanged, but retrieval and the relevance gate are bypassed.

Relevance gate (system).

You are screening retrieved passages for a difference-in-differences (DID) identification audit. For ONE identification dimension, decide whether each passage is actually evidence about THAT dimension’s assumption. This is a relevance check, NOT a quality or risk judgement. Mark a passage ‘high’ if it directly concerns the assumption (e.g. an event-study / pre-trend discussion for parallel trends), ‘partial’ if it touches it indirectly, and ‘none’ if it is about something else. Be strict: a passage that merely shares vocabulary but is about a different topic is ‘none’.

Relevance gate (user).

```

Identification dimension: <name>
Assumption: <assumption>
Testable implication: <implication>

Candidate passages:
[0] (<source>) <passage text, first 1200
characters>
[1] (<source>) <...>

Classify each passage's relevance to
this dimension.

```

Schema: one {index, relevance, reason} object per passage, with relevance in {high, partial, none}.

Adequacy judge (system).

You audit the causal-IDENTIFICATION credibility of difference-in-differences (DID) studies in environmental policy evaluation. You do NOT judge whether the estimated effect is true -- the counterfactual is never observed. For ONE identification dimension, judge whether the evidence the paper reports is ADEQUATE to support the assumption.

Crucial distinction: evidence PRESENCE is not evidence ADEQUACY. The mere mention of a topic (an event study, a balance table, 'placebo') does not make the assumption supported -- assess whether the reported evidence actually establishes the testable implication, and whether it reveals a flaw.

Return a risk level and an evidence-status label:

```

risk: low | medium | high
evidence_status: sufficient | partial
| missing | flawed
Map roughly: sufficient->low,
partial->medium, missing->high,
flawed->high. Be conservative: if the
evidence is absent, thin, or describes a
flawed design, do not call it supported.
Cite the specific evidence (by source)
your judgement rests on. Keep the
rationale to one or two sentences.

```

Adequacy judge (user).

```

Identification dimension: <name>
Assumption: <assumption>
Testable implication: <implication>
Evidence a credible paper would report:
<expected evidence>

```

```

Evidence retrieved from THIS paper:
- [<source>] <passage text>

```

Judge the adequacy of this evidence for the assumption.

When the gate keeps nothing the evidence block reads (no evidence was retrieved

for this dimension), but that path is unreachable in the deployed pipeline: a failed gate short-circuits to unknown without a model call (§5.3).

Schema: {risk, evidence_status, rationale, cited_evidence}.

Single-pass ablation, synthetic benchmarks (system and user).

You audit the causal-identification credibility of a difference-in-differences (DID) study. You do NOT judge whether the estimated effect is true. For EACH of the eleven identification dimensions listed, judge whether the evidence the paper reports is ADEQUATE to support that assumption. Evidence presence is not evidence adequacy. Return one judgement per dimension.

```

Identification dimensions to judge:
- <dimension id>: <assumption> (expect:
<expected evidence>)
[... eleven lines ...]

```

```

THE PAPER:
[<source>]
<full section text>

```

Judge the adequacy of the reported evidence for EVERY dimension.

Schema: a judgements array of {dimension, risk, evidence_status, rationale}, with dimension restricted to the eleven rubric ids and risk to low | medium | high; this arm cannot answer unknown. It produced the single-pass rows of Tables 1 and 4.

Single-pass variant, real-paper run (system, abridged to the risk block).

```

For each dimension output one risk
level:
low : adequate, credible evidence is
reported.
medium : evidence is partial,
indirect, or weak.
high : the required evidence is
absent, or present but flawed.
unknown : the text gives no basis to
judge either way (use sparingly).

```

Also output evidence_status in {sufficient, partial, missing, flawed} (roughly sufficient->low, partial->medium, missing/flawed->high). Be conservative: absent, thin, or flawed evidence is not 'supported'.

The real-paper contrast of Appendix C used this second implementation (`single_pass_assessor.py`). Its preamble restates the adequacy standard of the judge prompt above, it receives the whole paper with the rubric, and it returns free JSON keyed by dimension id in which a missing or malformed cell is coerced to unknown.

F Reproducibility Details

Code, configuration, prompts, committed run files, the frozen pilot inputs, the annotation protocol and the locked gold are available at <https://github.com/yonghongzhang-io/ARGUS>.

All gpt-4o stages use temperature 0 with `max_steps=1`; newer reasoning models (Claude Opus 4.8) reject an explicit temperature and run at provider defaults. The identification rubric, flaw taxonomy, and 33-variant catalogue are declarative configuration files in the code base, and the keyword baseline plus all leak-freedom (non-circularity) checks are deterministic and run without model access. Eleven-flaw numbers are July-2026 runs of the current runner; detection replicates exactly across the gpt-4o alias and the dated gpt-4o-2024-11-20 snapshot (localization varies between runs, 0.455–0.727, and the snapshot run rates one clean dimension medium, a false alarm of 0.09 where the reported run has none). A June-2026 run of the same code had scored two-stage 1.000/0.909 and single-pass 0.455; neither value was reproduced in July on the alias *or* the dated snapshot. The available records do not establish the cause of this difference, so we report the later runs with their dates and model identifiers. The 33-variant benchmark was run three times end-to-end (roughly 750 model calls per run, ~20 minutes and ~US\$8 per gpt-4o run; the local Llama run took ~67 minutes on a laptop at zero API cost); binary detection agreed on 32/33 variants across the three runs and target-dimension risk labels on 31/33, while the clean audit flagged a different dimension in run 1 than in runs 2 and 3, and Table 2 reports the three-run mean. The two-call-per-dimension pipeline audits one paper for roughly US\$0.25 of gpt-4o usage.

G Uncertainty Quantification

Table 8 gives Wilson 95% intervals for the main proportions reported in the main text and exact

tests between arms evaluated on the same items; Table 9 gives cell-bootstrap intervals (5,000 resamples) for the human-gold agreement metrics. At $n=11$ the keyword-versus-two-stage gap is marginal under the paired test on the same flaws (7 vs 1 discordant, $p=0.07$; an unpaired Fisher test, which ignores the pairing, gives $p=0.03$); at $n=33$ no pairwise difference between the single-pass, per-dimension, two-stage, and oracle arms reaches significance. For the calibration lift the comparison is paired, so overlap of the intervals in Table 9 is not the test. On the 33 answered cells the weak-retrieval rule alone turns seventeen wrong cells exact and none the other way (exact McNemar $p<0.001$); all four rules turn twenty ($p<0.001$). Those tests treat cells as independent, yet the cells are nested in five papers. By paper, the single rule improves exact agreement in all five papers, as do the four rules (sign test $p=0.0625$ in each case, the smallest value five papers allow). Neither analysis removes the bias of deriving the rules on the pilot they are scored on, and both count agreement with a gold that is 84% *medium*, which a rule that demotes to *medium* matches by construction. We therefore claim a consistent in-sample direction, not a demonstrated out-of-sample gain.

Tables 8 and 9 are computed from committed files by `uncertainty.py` and `calibration_recheck.py` in `experiments/ablations/`; the pilot cells they read are frozen with checksums in `experiments/annotation/pilot_frozen/` and the locked gold in `experiments/annotation/gold_final/`.

H The Real-Paper Corpus

The 26 papers audited in §5.3 are those tagged difference-in-differences in the corpus of 259 economics papers released with CausalVerify (Zhang et al., 2026a) (Table 10; titles in Table 11). The tag is the corpus’s method-family label, which we did not verify paper by paper, and it is imperfect: one entry is a literature review (below) and one identifies its effect from sibling comparisons. Nineteen entries are journal articles, one of them in a medical journal, and seven are held as National Bureau of Economic Research (NBER) working-paper versions. That corpus as released contains one byte-identical duplicate (a 2002 Clean Air Act study under two identifiers); both copies re-

Table 8: Wilson 95% confidence intervals (CI) for reported rates (top) and exact tests between arms (bottom; McNemar tests are paired on the same flaws or variants; discordant = detected by the first arm only / by the second only). False-alarm rows give counts only: every variant is paired with the same clean fixture, so they are repeated readings of one clean audit of eleven dimensions, not independent trials, and an interval would overstate what they show. Detection intervals describe this fixed test set (33 variants nested in eleven dimensions of one fixture), not real papers at large.

benchmark	quantity	k/n	95% CI
11 flaws	keyword detection	2/11	0.05–0.48
	single-pass detection	9/11	0.52–0.95
	per-dimension detection	11/11	0.74–1.00
	two-stage detection	8/11	0.43–0.90
	two-stage false alarm	0/11	–
33 variants	single-pass detection	24/33	0.56–0.85
	per-dimension detection	29/33	0.73–0.95
	two-stage detection (run 1)	25/33	0.59–0.87
	commission	20/22	0.72–0.97
	omission	5/11	0.21–0.72
	two-stage false alarm	3/33	–
	oracle-retrieval detection	30/33	0.76–0.97
cross-model	Opus 4.8 detection	30/33	0.76–0.97
	Opus 4.8 false alarm	27/33	–
	Gemini 2.5 Flash detection	26/33	0.62–0.89
	Gemini 2.5 Flash false alarm	6/33	–
	Llama 3.1 8B detection	31/33	0.80–0.98
	Llama 3.1 8B false alarm	9/33	–
test	arms	discordant	p
Fisher	keyword vs two-stage (11)	–	0.030
McNemar	keyword vs two-stage (11)	1 / 7	0.070
McNemar	single-pass vs per-dim. (33)	1 / 6	0.125
McNemar	per-dim. vs two-stage (33)	5 / 1	0.219
McNemar	single-pass vs two-stage (33)	5 / 6	1.000
McNemar	two-stage vs oracle (33)	1 / 6	0.125

ceived identical audits on all eleven dimensions from both assessors, and we count the paper once, so every count reads 26 papers and 286 cells. One further entry is a review of the school-spending literature rather than a primary DID study; we retain and note it. By topic, the corpus spans finance and banking (5), health (4), education (4), environment and energy (4), international trade (3), labour (3), crime and social policy (2), and industrial location (1); the environment-and-energy subset covers the industrial-activity effects of environmental regulation, air quality and early-life mortality, nonlinear electricity pricing, and gasoline prices and vehicle choice. None of the 26 is a climate-policy evaluation in the narrow sense (a carbon price, an emissions trading scheme, or an emission standard): the real corpus is a general-economics stress test of the audit architecture, and the climate-specific content of the paper is the rubric instantiation of

Table 9: Cell-bootstrap 95% intervals for agreement with the reconciled gold (answered cells only; quadratic-weighted κ ; 5,000 resamples). Rule 1 is the weak-retrieval rule; rules 1–4 add the three single-cell rules of Appendix J. Resampling cells ignores their nesting in five papers and may understate uncertainty.

policy	n	exact agreement	weighted κ
before	33	0.24 (0.09–0.39)	0.08 (–0.05–0.25)
rule 1, demote→med	33	0.76 (0.61–0.88)	0.23 (–0.22–0.63)
rule 1, abstain→unk	13	0.62 (0.38–0.85)	0.29 (–0.18–0.77)
rules 1–4, demote→med	33	0.85 (0.73–0.97)	0.22 (–0.42–0.79)
rules 1–4, abstain→unk	13	0.85 (0.62–1.00)	0.32 (–0.52–1.00)

Table 10: Venue of the 26 audited papers, as recorded for the version held in the source corpus. Two entries carry a corrupted venue field in the corpus index; they are counted under the journal that published them (Journal of Political Economy, 2003; Review of Economic Studies, 2002). AEJ: American Economic Journal; JF: Journal of Finance; REStat: Review of Economics and Statistics; REStud: Review of Economic Studies.

venue / version	papers
American Economic Review	4
Quarterly Journal of Economics	3
Journal of Political Economy	2
Journal of Financial Economics	2
AEJ: Applied Economics	2
JF / REStat / REStud / J. Monetary Econ. / J. Int. Econ. (one each)	5
New England Journal of Medicine	1
NBER working-paper versions	7

§3 and Appendix A; the synthetic fixture of §4 is a generic environmental-policy study (a China emissions-trading fixture ships with the code but produced none of the numbers reported here). The environment-and-energy subset has an unknown share of 0.43 (44 cells), against 0.39 in the remaining papers (242 cells); the corresponding high-risk shares are 0.34 and 0.46. With four papers in the subset, this comparison does not establish whether audit behaviour differs by domain.

I Qualitative Audit Examples

A complete audit report. Table 12 shows ARGUS’s actual output on the synthetic fixture with the *anticipation* flaw injected (§4): the report flags exactly the damaged dimension as high, giving the anticipation-gap rationale, keeps nine supported dimensions at low, and shows a mild over-severity trace on *inference* (the fixture does not state the number of clusters), the same bias the human-gold pilot quantifies. This is the dimension-level, evidence-cited report a reviewer receives.

Table 11: The 26 audited studies, by title as indexed in the source corpus (the duplicated entry is listed once).

The Effect of the Banking Panic on the Supply of Credit to the Corporate Sector
The Effects of State-Level Banking Competition on Innovation
Do Credit Market Shocks Affect the Real Economy? Quasi-Experimental Evidence from the Great Recession and ‘Normal’ Economic Times
Cyclicality of Credit Supply: Firm Level Evidence
Does Stock Liquidity Enhance or Impede Firm Innovation?
Minimum Wages and Employment: A Case Study of the Fast Food Industry in New Jersey and Pennsylvania
Minimum Wage Effects Across State Borders: Estimates Using Contiguous Counties
Hospital Ownership and Public Medical Spending
The Impact of Public Disclosure of Health Outcomes on Patient and Social Welfare
Did Community Rating Induce an Adverse Selection Death Spiral? Evidence from New York, Pennsylvania, and Connecticut
Mortality and Access to Care among Adults after State Medicaid Expansions
Early Childhood Intervention and Life-Cycle Skill Development: Evidence from Head Start
The Impacts of Environmental Regulations on Industrial Activity: Evidence from the 1970 and 1977 Clean Air Act Amendments and the Census of Manufactures
Legalized Abortion and Crime
Do Consumers Respond to Marginal or Average Price? Evidence from Nonlinear Electricity Pricing
Estimates of the Impact of Crime Risk on Property Values from Megan’s Laws
Effects of Liberalized Trade on Plant Productivity
The Surprisingly Swift Decline of U.S. Manufacturing Employment
Factor Immobility and Trade: Poverty in India
Trade Liberalization and Industry Wage Premia: Evidence from Colombia
The Effect of State Policies on the Location of Manufacturing: Evidence from State Borders
Air Quality and Early-Life Mortality: Evidence from Indonesia’s Wildfires
Does School Spending Matter? The New Literature on an Old Question
The Evolution of Black/White Earnings: 1960–1980
The Effects of Class Size on Student Achievement: New Evidence from Population Variation
Are Consumers Myopic? Evidence from New and Used Car Purchases

A grounded judgement on a real paper.

On a trade-and-wages paper ARGUS retrieves the employment-results section and flags *inference* as high because “the paper only mentions heteroskedastic-consistent standard errors and does not provide information on clustering or the number of clusters”: a specific econometric observation, not a keyword hit. Elsewhere it distinguishes a *related but different* reported check (an alternative-specification section) from the placebo evidence the rubric specifies for that dimension.

J Supplementary Figures and Tables

Pilot-gold construction (§5.4). The protocol and analysis plan were written and committed before any label was returned (`annotation/human_pilot/PROTOCOL.md`); each returned workbook is recorded by SHA-256 on receipt, and the gold is locked once (`experiments/`

Table 12: ARGUS audit report (abridged) on the fixture with the injected anticipation flaw. Rationales are abridged from the evidence-cited report; the injected dimension is the one flagged high.

dimension	risk	rationale (abridged)
no anticipation	high	outcomes began changing between the policy announcement and its rollout, undermining no-anticipation
inference	medium	clustering and wild-cluster bootstrap reported, but the number of clusters is not stated
parallel trends	low	event-study with null pre-period coefficients and placebo tests
staggered timing	low	heterogeneity-robust estimators with a Goodman-Bacon decomposition
7 further dimensions	low	supporting evidence retrieved and judged adequate

`annotation/gold_final/`). The two annotators, a student at the first author’s university and a member of a vocational college in China, neither an author nor involved in building ARGUS, read the five PDFs themselves, used no AI tool, saw no ARGUS output, did not confer before returning, and signed to that effect in the workbook. The five papers are `paper_01`, `03`, `07`, `08` and `10` of the repository’s corpus manifest, chosen in June 2026 before any label existed. Independent labels: 55 cells each; applicability agrees on 49, with one cell rated not applicable by one annotator; of the 54 cells both rated, 45 agree exactly (Cohen’s $\kappa = 0.56$; quadratic-weighted $\kappa = 0.62$), and all nine risk disagreements are a single severity step apart. The 13 cells that differed in applicability or risk were settled by the two annotators together, all 13 as agreed decisions, 9 of them under an analogue reading of the dimension; no third person was needed. Each annotator’s verbatim quotes were checked against the PDFs: 50/55 and 49/55 occur verbatim, the rest as partial matches, one cell each not found. Three deviations are logged: one annotator first returned a paper sheet pasted from another paper and corrected it herself within sixteen minutes; the file first recorded for her had been re-saved on the first author’s computer (how is not known) and differed from her own last version in eleven evidence-location cells and one rationale, in no label, quote or confidence, so the record was rebuilt from her own file and none of the twelve cells enters the analysis; and the other annotator first returned the workbook without the sign-off sheet and re-sent it signed.

Change from the reviewed version. The reviewed version described two annotators and an adjudication round. No record of how those la-

bels were produced survives: the first author’s recollection is that two people produced them with LLM assistance and sent them to the first author, who entered them; there is no file, sign-off or adjudication record of theirs, so the reviewed description was not supported by any record, and the labels are not treated as an annotation. They are kept, checksummed, in `experiments/annotation/pilot_frozen/` and are replaced here. Against the human gold they agree on 31/55 cells (Cohen’s $\kappa = 0.23$), being less severe on 23 (low where the gold is medium); under them ARGUS was more severe on 28/33 answered cells and exact on 5, against 25/33 and 8 now, and the reviewed calibration figures (exact agreement $0.12 \rightarrow 0.36, 0.45$ with four rules) are superseded by §5.6: the larger lift now reflects the medium-heavy gold, not the rule.

In the leave-one-paper-out check (§5.6), the demotion rule’s trigger conditions recur in all five folds: among each fold’s over-severe *high* cells, 79–86% have weak retrieval and 94–100% have missing evidence (computed on the rich re-run of §5.6). The folds test whether the error pattern recurs; no rule was refitted on four papers and scored on the fifth.

The calibration rules. The layer acts on *high* cells only and never raises a risk; the first matching rule applies. Rule 1 (20 cells): `retrieval_quality=weak` $\rightarrow$ *medium*, or $\rightarrow$ *unknown* under the abstain policy. Three further rules each fire on one cell and demote to *medium*: rule 2, the dimension is staggered timing and the rationale names Callaway, Sun–Abraham, Goodman–Bacon, or “modern estimator”; rule 3, the dimension is inference, `evidence_status=missing`, and the rationale mentions “cluster” or “robust standard errors”; rule 4, the dimension is concurrent policies and `retrieval_quality=good`. Rules 2–4 are post-hoc string and field rules defined after inspecting the errors on this same pilot, and all three cells they touch land on the gold label. They are not principled: naming an estimator, or the Goodman–Bacon decomposition (a diagnostic of two-way fixed-effects comparisons, not a heterogeneity-robust estimator), does not show that it was used appropriately; a rationale that mentions clustering may be reporting its absence; and rule 4 has no content condition at all (locating relevant evidence is not the evidence sup-

Table 13: Two-stage ARGUS vs reconciled human gold (55 cells, 5 papers) under two unknown policies: coverage (drop abstentions) and strict (unknown counts as mismatch). Pilot-scale: only 2 gold cells are *high*; the robust signal is the one-directional over-severity and the abstention rate.

unknown policy	<i>n</i>	exact	κ	wt. κ
exclude (coverage)	33	0.242	0.031	0.128
mismatch (strict)	55	0.145	0.007	–

Table 14: Deterministic calibration vs the human gold (55 cells), in-sample. Two policies for a weak-retrieval *high*: demote to *medium* (med, keep coverage) or abstain to *unknown* (unk). Rule 1 alone is the result we claim; rules 1–4 add three single-cell exploratory rules (Appendix J). Under abstention rule 1 changes no retained label: its higher agreement comes from dropping 20 answers. Pilot scale: only 2 gold cells are *high*; a constant *medium* label scores 26/33 on the answered cells, so exact agreement carries little information here. The before column is the rich re-run of §5.6, which differs from the corpus run of Table 13 in one cell (a gold-low cell moves from *medium* to *high*, which is why weighted κ is 0.08 here and 0.13 there); intervals in Table 9.

metric	before	rule 1		rules 1–4	
		med	unk	med	unk
over-severe (answ.)	25	8	5	5	2
high precision	1/25	1/5	1/5	1/2	1/2
high recall	1/2	1/2	1/2	1/2	1/2
exact agr. (answ.)	0.24	0.76	0.62	0.85	0.85
wt. κ (answ.)	0.08	0.23	0.29	0.22	0.32
answered / unk	33/22	33/22	13/42	33/22	13/42

porting the assumption). We treat only rule 1 as a result. The script (`calibrate_argus_v1_pilot.py`) reproduces the historical outputs on all 55 cells.

The remaining material is Figure 5 (per-dimension risk on the real corpus) and Tables 13–14 with Figure 6 (agreement and calibration against the gold).

K Shared-Evidence Control

Table 1 compares pipelines that differ in the judgement policy *and* in the evidence they retrieve. This control holds the evidence fixed. Its design, thresholds, and decision rule were committed before this control was run (SHARED_EVIDENCE_PROTOCOL.md, commit 288e01f, 20 September 2026, tagged in the repository; the runner, the two zero-call cells, and the model calls followed). Two evidence bundles are frozen to disk for every

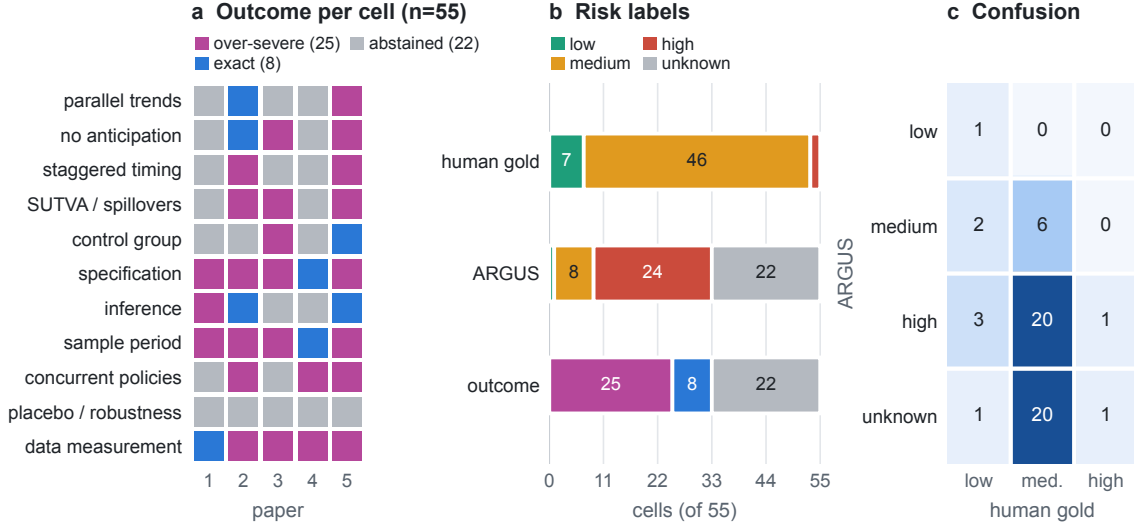


Figure 4: Two-stage ARGUS versus the reconciled gold on the 55 pilot cells (frozen protocol). **(a)** Every cell, eleven dimensions by five papers, coloured by outcome: ARGUS abstains on 22 and, of the 33 it answers, is more severe than the gold on 25, matches it on 8, and is less severe on none. **(b)** Distribution of risk labels in the gold and in ARGUS’s output, and the outcome of the 55 cells in the colours of panel (a). **(c)** ARGUS-by-gold confusion matrix (counts).

Table 15: Shared-evidence control: evidence bundle (K, keyword chunks; S, ungated sections) $\times$ judgement policy (kw, keyword scorer; llm, adequacy judge). det / fa / loc as in Table 4; off = share of the ten non-target dimensions rated $\geq$ medium on the injected variants. The reference row is the gated pipeline on the gpt-4o alias (detection and false alarm from run 1, Table 8; localization is the three-run mean of Table 2).

cell	11 flaws			33 variants			
	det	fa	loc	det	fa	loc	off
K-kw	2/11	0/11	2/11	8/33	0/33	8/33	0.00
K-llm	10/11	1/11	8/11	31/33	3/33	22/33	0.15
S-kw	1/11	0/11	1/11	2/33	0/33	2/33	0.00
S-llm	11/11	0/11	11/11	31/33	0/33	22/33	0.16
<i>gated two-stage</i>	8/11	0/11	8/11	25/33	3/33	0.66	–

paper and dimension: K, the keyword pipeline’s stage-2 chunks, and S, the LLM path’s four section candidates with the relevance gate switched off. Each bundle is read by both policies, the keyword scorer and the adequacy judge (gpt-4o-2024-11-20, temperature 0, prompts of Appendix E verbatim). No cell is gated, so none can abstain. All 45 papers (one clean fixture, 11 flaws, 33 variants) are judged on all eleven dimensions: 990 adequacy calls, none failed. Cell K-kw reproduces the published keyword baseline exactly.

Result. With the evidence identical, the policy contrast is large under both bundles (Table 15): on K, detection rises from 2 to 10 of 11 flaws (eight gained, none lost; exact McNemar $p=0.008$) and

from 8 to 31 of 33 variants ($p<0.001$); on S, from 1 to 11 and from 2 to 31. The evidence contrast at a fixed judge is small (10 vs 11 of 11; 31 vs 31 of 33). The pre-specified rule asked for a gain of at least 0.30 on the 11 flaws, the same direction on the variants, and false-alarm and clean off-target rates within 0.10 of the keyword scorer’s. All three hold, the last narrowly (0.09), so the policy explanation is supported: on these fixtures the judgement policy, not the retrieval path, carries the gap in Table 1.

What it costs, and what it does not show. The gain is a trade. The judge rates 11–16% of non-target dimensions medium or higher on injected papers, where the keyword scorer flags none. On the keyword chunks the judge also rates one clean dimension (SUTVA) medium. Every false alarm in the K-llm row is that single verdict, counted once per flaw that targets the dimension: both benchmarks share one clean fixture, so the false-alarm rates here and in Tables 1 and 2 rest on one clean audit of eleven dimensions, not on 11 or 33 independent trials. The ungated S-llm cell detects more than the gated pipeline (omission variants 11/11 vs 5/11), the trade §5.2 describes: shown four sections of a short fixture, the judge notices the deleted one, where the gate abstains. The reference row was run on the gpt-4o alias, so that comparison is indicative; fixtures, sentinel lists, and injector are co-designed, and n is small.

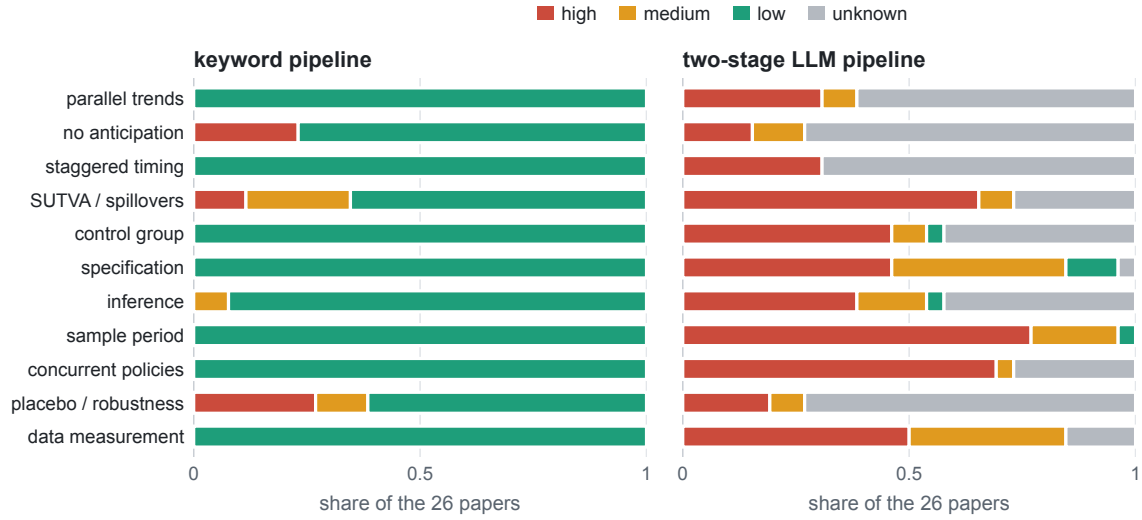


Figure 5: Per-dimension risk distribution on the 26 papers tagged DID in the source corpus. The keyword baseline (left) is almost entirely *low*; the two-stage LLM (right) is differentiated, with a large *unknown* share that marks retrieval failure rather than over-flagging.

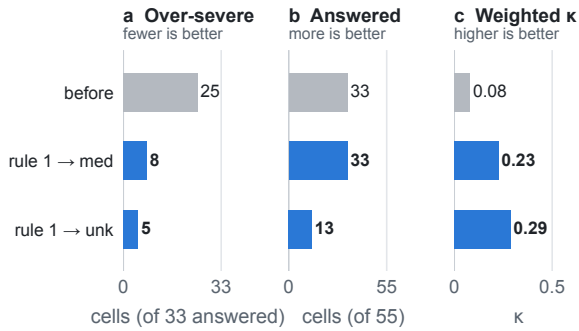


Figure 6: **Calibration trade-off: severity reduction versus coverage** against the reconciled human gold, for the weak-retrieval rule alone (in-sample; med and unk are the two policies of Table 14, which also gives the exploratory four-rule set). Demoting weak-retrieval *high*s to *medium* preserves coverage (33 answered) while cutting over-severe judgements from 25 to 8; abstaining leaves fewer over-severe cells (5) and a higher weighted κ (0.29) only by discarding 20 answers, and leaves the remaining labels unchanged. Most of the gain in exact agreement reflects a gold that is 84% *medium* (§5.6); the remaining bottleneck is evidence grounding, not severity thresholding alone.